



AMITY UNIVERSITY
UTTAR PRADESH

AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

PROJECT PROGRESS REPORT

B. Tech (Computer Science and Engineering)

Group No: 50

Project Title: GuardShield AI: Real-Time Sidecar Proxy for LLM Hallucination Detection & Prompt Injection Defense

Area: Generative AI & AI Safety (Trustworthy AI / MLOps)

Academic Session: 2026-2027

Project Guide: Dr. Abhishek Kaushal

Details of Project Team:

Programme:- B.TECH CSE		Year/Semester:- 4th Year / 7th Semester	
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Paper Summary & Literature Review:

Introduction

As Generative Large Language Models (LLMs) like GPT-4, Claude 3.5, and Llama 3.1 are deployed into production enterprise applications (such as medical assistants, legal assistants, and banking chatbots), two critical vulnerabilities pose severe operational risks: (1) Adversarial Prompt Injections & Jailbreaking attacks that trick LLMs into violating security policies or leaking confidential system prompts, and (2) Model Hallucinations where LLMs generate non-factual, unsupported assertions with high confidence.

Existing commercial guardrails (e.g., Meta Llama-Guard-3 or offline NLI fact-checkers like SelfCheckGPT) suffer from severe latency bottlenecks, adding 1.5 to 3.0 seconds of delay per query or evaluating responses only after full text generation completes—making real-time streaming user experiences unfeasible. To address these fundamental limitations, GuardShield AI introduces a high-

performance, dual-stage asynchronous sidecar proxy designed to inspect inputs and output streaming tokens with sub-25ms latency overhead.

1. Adversarial Prompt Injection & Jailbreaking Benchmarks

The vulnerability of LLMs to adversarial prompt injections has been extensively investigated. Chao et al. [1] introduced JailbreakBench (NeurIPS 2024), establishing a standardized robustness evaluation framework for jailbreaking attacks across closed and open LLMs. Schulhoff et al. [2] conducted a global study with HackAPrompt (EMNLP 2023), demonstrating that simple linguistic obfuscations and system prompt override sequences reliably compromise commercial model alignments. Ji et al. [3] released BeaverTails (NeurIPS 2023), constructing a multi-dimensional preference dataset separating harmful queries from benign intents across 14 safety categories. Zhang et al. [4] presented CAPTURE (ACL 2025), emphasizing that context-aware prompt injection guardrails must evaluate intent semantics rather than simplistic keyword blocklists. Inan et al. [5] proposed Llama-Guard (Meta AI 2024), a safety classifier based on an 8B parameter LLM; however, its substantial computational requirements restrict its utility as a sub-25ms microservice proxy on commodity hardware.

2. LLM Hallucination Detection & Factual Verification

Hallucination detection in generative models has evolved from offline sampling to factual entailment verification. Manakul et al. [6] proposed SelfCheckGPT (EMNLP 2023), a zero-resource hallucination detection framework using multiple stochastic sample comparisons; however, generating multiple auxiliary samples introduces significant compute overhead and multi-second latency lags. Li et al. [7] formulated HaluEval (EMNLP 2023), a comprehensive benchmark for evaluating hallucinations across question-answering, dialogue, and summarization tasks. Lin et al. [8] introduced TruthfulQA (ACL 2022), demonstrating that autoregressive LLMs frequently mimic human falsehoods and require external verification. Thorne et al. [9] developed the FEVER benchmark (NAACL 2018), standardizing fact extraction and Natural Language Inference (NLI) entailment scoring against verified evidence corpora.

3. Small Language Models & Efficient Representation Learning

Deploying guardrails within tight latency constraints requires compact Small Language Models (SLMs). Devlin et al. [10] introduced BERT, establishing masked bidirectional transformer pre-training. He et al. [11] developed DeBERTa-v3, incorporating disentangled attention and enhanced masked language modeling with ELECTRA-style training, significantly outperforming RoBERTa [12] and traditional BERT architectures at matching parameter scales. Reimers and Gurevych [13] introduced Sentence-BERT and Cross-Encoders, demonstrating that cross-attention between premise and hypothesis yields superior Natural Language Inference (NLI) classification accuracy compared to bi-encoder cosine similarity. Sanh et al. [14] introduced DistilBERT, validating that knowledge distillation enables 40% model size reduction with minimal performance degradation.

4. Real-Time Token Logit Entropy & Streaming Inspection

Monitoring model generation during token streaming provides a viable pathway to eliminate offline latency bottlenecks. Shannon [15] formulated mathematical entropy over discrete probability distributions. Azaria and Mitchell [16] demonstrated that internal representation logits and token output distributions contain measurable signals of factual veracity. Kuhn et al. [17] introduced

Semantic Entropy, proving that semantic dispersion across token clusters correlates strongly with hallucination likelihood. Kadavath et al. [18] showed that language models can express calibrated probabilities regarding their own output correctness. Varshney et al. [19] investigated confidence scores and token-level uncertainty for early termination in conversational systems.

5. Synthesis for the Proposed GuardShield AI Framework

Across the literature, a clear architectural consensus emerges: robust LLM safety requires both Pre-Scan input filtration and Post-Scan output monitoring. GuardShield AI synthesizes these insights into an asynchronous dual-stage sidecar proxy architecture. In the Pre-Scan stage, a fine-tuned microsoft/deberta-v3-small (86M parameters) multi-class classifier inspects incoming user prompts with sub-20ms latency overhead on local GPU hardware (NVIDIA RTX 3050 / Apple Silicon MPS), blocking prompt injections and system exfiltration attempts prior to reaching the target LLM. In the Post-Scan stage, as the local Ollama LLM (Llama-3.1-8B) streams response tokens back to the client, GuardShield continuously monitors Shannon Entropy over token logit distributions: $H(x) = -\sum(p * \log_2(p))$. When entropy spikes past a calibrated threshold, an asynchronous windowed NLI Cross-Encoder evaluates factual entailment mid-sentence, triggering early token termination before hallucinated content reaches the user.

Conclusion

The literature reviewed in this progress report validates the core hypotheses of GuardShield AI: (1) An 86M parameter DeBERTa-v3 SLM provides competitive prompt injection classification accuracy while reducing compute overhead by over 98% compared to 8B parameter alternatives (Llama-Guard-3), and (2) Real-time token logit entropy monitoring resolves the multi-second offline latency penalty of existing hallucination detectors (SelfCheckGPT). Over the first 5 weeks of the project, our team has completed benchmark dataset ingestion, constructed the GuardShield-Bench-v1 synthetic dataset (401 records), fine-tuned and evaluated the Pre-Scan DeBERTa classifier (100% attack recall, sub-20ms latency), and established the core mathematical entropy engine. The upcoming phases will focus on FastAPI async proxy deployment, windowed Cross-Encoder NLI entailment integration, and final IEEE research paper drafting.

PERT Chart / Schedule of Project Completion:-

[illegible]

Benchmark Dataset Ingestion (JailbreakBench, BeaverTails, TruthfulQA)		✓	✓									
GuardShield- Bench-v1 Synthetic Dataset Creation (Base64, Hinglish)			✓	✓								
Pre-Scan DeBERTa-v3 Classifier Fine- Tuning & GPU Checkpointing				✓	✓							
Pre-Scan Evaluation, Recall Benchmarking & Sub-25ms Latency Profiling					✓	✓						
Streaming Token Logit Entropy Module & Early Termination Logic						✓	✓	✓				
Windowed Cross-Encoder NLI Entailment Integration							✓	✓	✓			
FastAPI Async								✓	✓	✓		

SSE Sidecar Proxy Server Development												
Comparative Benchmarking vs Llama- Guard-3 & SelfCheckGPT									✓	✓	✓	
Final IEEE Research Paper Compilation, Synopsis & Viva Demo											✓	✓

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